1(a)	$2x^3-4x-15=(Ax+B)(x^2+3x+4)+C(2x+3)\Rightarrow A=...$ or $x^2+3x+4\overline{)2x^3+0x-4x-15}$ $\quad 2x...$ $\quad 2x^3+...$	**M1**
	$A=2$	**A1**
	$B=-6, C=3$	**A1A1**
		(4)
(b)	$\int f(x)\,dx="A"\frac{x^2}{2}+"B"x+"C"\ln\left(x^2+3x+4\right)$	**M1** **A1ft**
	$\int f(x)\,dx=x^2-6x+3\ln\left(x^2+3x+4\right)$	
	$\int_3^5 f(x)\,dx=\left(5^2-6\times5+3\ln\left(5^2+3\times5+4\right)\right)-\left(3^2-6\times3+3\ln\left(3^2+3\times3+4\right)\right)=..$	**dM1**
	$=-5+3\ln 44+9-3\ln 22=4+3\ln\left(\frac{44}{22}\right)$	**M1**
	$=4+\ln 8$	**A1**
		(5)

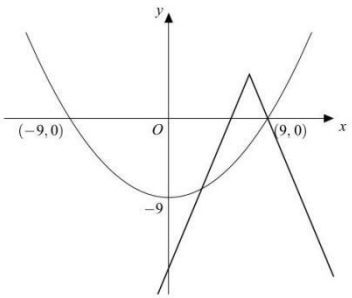
2(a)	Either $\mathrm{f}(x) < 5$ or $\mathrm{f}(x) \ldots 3$	**M1**
	$3,,\ \mathrm{f}(x) < 5$	**A1**
		(2)
(b)	(i) $y = 5 - \dfrac{4}{3x+2} \Rightarrow \dfrac{4}{3x+2} = 5 - y \Rightarrow \dfrac{4}{5-y} = 3x + 2 \Rightarrow x = \ldots$	**M1**
	$\left(\mathrm{f}^{-1}(x)\right) = \dfrac{1}{3}\left(\dfrac{4}{5-x} - 2\right)$ oe such as $\dfrac{4}{15-3x} - \dfrac{2}{3}$ or $\dfrac{2x-6}{15-3x}$	**A1**
	(ii) Domain is $3,,\ x < 5$	**B1ft**
		(3)
(c)	$\mathrm{fg}(-\pi) = \mathrm{f}\left(\left\lvert 4\sin\left(-\dfrac{\pi}{3} + \dfrac{\pi}{6}\right)\right\rvert\right) = \mathrm{f}\left(\left\lvert 4\sin\left(-\dfrac{\pi}{6}\right)\right\rvert\right) = \mathrm{f}(2) = \ldots$	**M1**
	$= 5 - \dfrac{4}{6+2} = \dfrac{9}{2}$	**A1**
		(2)

3(a)	$f'(x) = 2(x-2) \times e^{3x} + (x-2)^2 \times 3e^{3x}$	**M1 A1**
	$f'(x) = 0 \Rightarrow e^{3x}(x-2)(2+3(x-2)) = 0 \Rightarrow x = ...$	**M1**
	$\Rightarrow x = \frac{4}{3}$	**A1**
	$y = \frac{4}{9}e^4 \qquad \left(\text{so } A \text{ is } \left(\frac{4}{3}, \frac{4}{9}e^4\right)\right)$	**A1**
		(5)
(b)	$k > 0$ or k ,, their $\frac{4}{9}e^4$ or with … or < respectively.	**M1**
	$0 < k$,, their $\frac{4}{9}e^4$	**A1ft**
		(2)

4(a)	$y = \log_{10}(2x+1) \Rightarrow 10^y = 2x+1 \Rightarrow x = ...$	**M1**
	$\Rightarrow x = \dfrac{10^y - 1}{2}$	**A1**
		(2)
(b)	$\left(\dfrac{\mathrm{d}x}{\mathrm{d}y} =\right) \dfrac{1}{2} 10^y \ln 10$	**M1**
	$\dfrac{\mathrm{d}y}{\mathrm{d}x} = 1 \Big/ \dfrac{\mathrm{d}x}{\mathrm{d}y} = \dfrac{1}{\frac{1}{2} 10^y \ln 10}$	**M1**
	$\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{2}{(2x+1)\ln 10}$	**A1**
		(3)

5(a)	$P(1)=\frac{4-1}{10}+\frac{3}{4}\ln\left(\frac{2}{3^2}\right)$ $(=-0.828...)$	**M1**
	$P(1)=-0.828...$ (which is negative so a loss of £0.828 million,) so approximately £830 000 loss.	**A1***
		(2)
(b)	$P(6)=-0.08799\ldots$ and $P(7)=0.1975\ldots$	**M1**
	There is a sign change and hence as P is continuous on [6,7], so the root for t lies in [6,7].	**A1**
		(2)
(c)	$P=0\Rightarrow\frac{4t-1}{10}=-\frac{3}{4}\ln\left(\frac{t+1}{(2t+1)^2}\right)\Rightarrow t=...$	**M1**
	$\Rightarrow t=\frac{10\left(-\frac{3}{4}\ln\left(\frac{t+1}{(2t+1)^2}\right)\right)+1}{4}=\frac{1}{4}+\frac{30}{16}\ln\left(\frac{t+1}{(2t+1)^2}\right)^{-1}$ $\Rightarrow t=\frac{1}{4}+\frac{15}{8}\ln\left(\frac{(2t+1)^2}{t+1}\right)$ *	**A1***
		(2)
(d)	$t_2=\frac{1}{4}+\frac{15}{8}\ln\left(\frac{13^2}{7}\right)=...(=6.219978...)$	**M1**
	$t_2=$ awrt 6.220	**A1**
	$t_6=6.314$	**A1**
		(3)
(e)	"6.3..."$\times 12=...$ months, or repeated iteration to root 6.31487 gives 75.7785… or allow "0.314..."$\times 12=$"3.768..."	**M1**
	So it will take 76 months. (Accept 75 or awrt 76 months)	**A1**
		(2)

6	$\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\cos x+\sin x\right)=-\sin x+\cos x$	**B1**
	$\dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{\left(\cos x+\sin x\right)\left(3\cos x\right)-\left(-\sin x+\cos x\right)\left(2+3\sin x\right)}{\left(\cos x+\sin x\right)^2}$	**M1** **A1**
	$=\dfrac{3\cos^2 x+\cancel{3\sin x\cos x}+2\sin x+3\sin^2 x-2\cos x-\cancel{3\sin x\cos x}}{\cos^2 x+2\sin x\cos x+\sin^2 x}$ $=\dfrac{3+2\sin x-2\cos x}{1+2\sin x\cos x}$	**M1**
	$=\dfrac{3+2\sin x-2\cos x}{1+2\sin x\cos x}\times\dfrac{\sec x}{\sec x}=\dfrac{3\sec x+2\sec x\sin x-2}{\sec x+2\sin x}$	**M1**
	$=\dfrac{2\tan x+3\sec x-2}{\sec x+2\sin x}$	**A1**
		(6)

7(a)	(i) $\left(\frac{22}{3}, 5\right)$	**B1**
	(ii) $(0, -17)$	**B1**
		(2)
(b)	$5-(3x-22)=0 \Rightarrow x=...$ **and** $5+(3x-22)=0 \Rightarrow x=...$	**M1**
	$x=9$ and $x=\frac{17}{3}$	**A1**
		(2)
(c)	 Correct U shape symmetric about *y*-axis with vertex on negative *y*-axis.	**B1**
	Graphs meet $(9,0)$ with $(-9,0)$ also shown.	**B1**
	Intercept at $(0,-9)$ stated or labelled.	**B1**
		(3)
(d)	Intersect at $(9,0)$	**B1**
	Or when $5+3x-22=\frac{1}{9}x^2-9$	**M1**
	$\Rightarrow (x-3)(x-24)=0 \Rightarrow x=...$	**dM1**
	Need smaller root $x=3 \Rightarrow y=...$	**dM1**
	$(3,-8)$	**A1**
		(5)

8(a)	$R = 17$	**B1**
	$\tan\alpha = \frac{15}{8}$	**M1**
	$\alpha = 1.081$	**A1**
		(3)
(b)	(i) Min f(x) $= \frac{15}{41 + 2\times"17"}$	**M1**
	$= \frac{1}{5}$	**A1**
	(ii) Occurs when $\sin(x - 1.081) = 1 \Rightarrow x - 1.081 = \frac{\pi}{2} \Rightarrow x = ...$	**M1**
	$x =$ awrt 2.65	**A1**
		(4)
(c)	$-\frac{23}{5}$ (or -4.6)	**B1ft**
		(1)
(d)	Awrt 1.33	**B1ft**
		(1)

9(a)	$\dfrac{\cos^2\theta}{\cos 2\theta - \sin 3\theta} = \dfrac{\cos^2\theta}{\cos 2\theta - (\sin 2\theta\cos\theta + \cos 2\theta\sin\theta)}$	**M1**
	$= \dfrac{\cos^2\theta}{1-2\sin^2\theta - 2\sin\theta\cos^2\theta - \sin\theta(1-2\sin^2\theta)}$	**M1**
	$= \dfrac{1-\sin^2\theta}{1-3\sin\theta-2\sin^2\theta+4\sin^3\theta} = \dfrac{(1-\sin\theta)(1+\sin\theta)}{(1-\sin\theta)(1-2\sin\theta-4\sin^2\theta)}$	**M1**
	$= \dfrac{1+\sin\theta}{1-2\sin\theta-4\sin^2\theta}$ *	**A1***
		(4)
(b)	$\dfrac{1+\sin\theta}{1-2\sin\theta-4\sin^2\theta} = 2\operatorname{cosec}\theta \Rightarrow (1+\sin\theta)\sin\theta = 2(1-2\sin\theta-4\sin^2\theta)$	**M1**
	$\Rightarrow 9\sin^2\theta + 5\sin\theta - 2 = 0$	**A1**
	$\Rightarrow \sin\theta = \dfrac{-5\pm\sqrt{25-4\times9\times-2}}{18} = \ldots \Rightarrow \theta = \sin^{-1}\ldots$	**M1**
	Two of $\theta = 15.6°, 164.4°, 235.6°, 304.4°$	**A1**
	All of $\theta = 15.6°, 164.4°, 235.6°, 304.4°$	**A1**
		(5)
9(a)	$\dfrac{\cos^2\theta}{\cos 2\theta - \sin 3\theta} = \dfrac{\cos^2\theta}{\cos 2\theta - (\sin 2\theta\cos\theta + \cos 2\theta\sin\theta)}$	**M1**

Example Alt	$= \dfrac{1-\sin^2\theta}{\cos 2\theta(1-\sin\theta)-2\sin\theta(1-\sin^2\theta)}$ $= \dfrac{(1-\sin\theta)(1+\sin\theta)}{(1-\sin\theta)\cos 2\theta-2\sin\theta(1+\sin\theta)(1-\sin\theta)}$	**3rd M1**
	$= \dfrac{(1-\sin\theta)(1+\sin\theta)}{(1-\sin\theta)(1-2\sin^2\theta-2\sin\theta-2\sin^2\theta)}$	**2nd M1**
	$= \dfrac{1+\sin\theta}{1-2\sin\theta-4\sin^2\theta}$ *	**A1***
		(4)
9(a) **Alt2** **In reverse**	$\dfrac{1+\sin\theta}{1-2\sin\theta-4\sin^2\theta}\times\dfrac{1-\sin\theta}{1-\sin\theta} = \dfrac{1-\sin^2\theta}{1-3\sin\theta-2\sin^2\theta+4\sin^3\theta}$ $= \dfrac{\cos^2\theta}{1-3\sin\theta-2\sin^2\theta+4\sin^3\theta}$	**3rd M1**
	$= \dfrac{\cos^2\theta}{1-2\sin^2\theta-3\sin\theta+4\sin^3\theta} = \dfrac{\cos^2\theta}{\cos 2\theta-(3\sin\theta-4\sin^3\theta)}$	**2nd M1**
	$= \dfrac{\cos^2\theta}{\cos 2\theta-\sin\theta(3-4\sin^2\theta)} = \dfrac{\cos^2\theta}{\cos 2\theta-\sin\theta(1+2(1-\sin^2\theta)-2\sin^2\theta)}$ $= \dfrac{\cos^2\theta}{\cos 2\theta-\sin\theta(1+2\cos^2\theta-(1-\cos 2\theta))}$ $= \dfrac{\cos^2\theta}{\cos 2\theta-(2\sin\theta\cos^2\theta+\sin\theta\cos 2\theta)}$ $= \dfrac{\cos^2\theta}{\cos 2\theta-(\sin 2\theta\cos\theta+\cos 2\theta\sin\theta)}$	**1st M1**
	$= \dfrac{\cos^2\theta}{\cos 2\theta-\sin 3\theta}$ *	**A1***
		(4)

9(a) **Alt3** **By cross multiplying**	$\dfrac{\cos^2\theta}{\cos 2\theta - \sin 3\theta} = \dfrac{1+\sin\theta}{1-2\sin\theta-4\sin^2\theta} \Leftrightarrow$ $\cos^2\theta\left(1-2\sin\theta-4\sin^2\theta\right) = \left(1+\sin\theta\right)\left(\cos 2\theta - \sin 3\theta\right) \Leftrightarrow$ $\cos^2\theta\left(1-2\sin\theta-4\sin^2\theta\right) = \left(1+\sin\theta\right)\left(\cos 2\theta - \underline{\sin 2\theta\cos\theta - \cos 2\theta\sin\theta}\right)$	**M1**
	$\Leftrightarrow \cos^2\theta\left(1-2\sin\theta-4\sin^2\theta\right)$ $= \left(1+\sin\theta\right)\left(\underline{1-2\sin^2\theta-2\sin\theta\cos^2\theta-\left(1-2\sin^2\theta\right)\sin\theta}\right)$	**M1**
	$\Leftrightarrow \cos^2\theta - \cancel{2\sin\theta\cos^2\theta} - 4\sin^2\theta\cos^2\theta = 1-2\sin^2\theta - \cancel{2\sin\theta\cos^2\theta}$ $- \cancel{\sin\theta} + \cancel{2\sin^3\theta} + \cancel{\sin\theta} \cancel{-2\sin^3\theta} - 2\sin^2\theta\cos^2\theta - \sin^2\theta + 2\sin^4\theta$ $\Leftrightarrow \cos^2\theta - 2\sin^2\theta\cos^2\theta = 1-3\sin^2\theta + 2\sin^2\theta\underline{\left(1-\cos^2\theta\right)}$	**M1**
	$\Leftrightarrow \cos^2\theta - \cancel{2\sin^2\theta\cos^2\theta} = 1-3\sin^2\theta + 2\sin^2\theta - \cancel{2\sin^2\theta\cos^2\theta}$ $\Leftrightarrow \cos^2\theta = 1-\sin^2\theta$ which is true, hence given result is true.*	**A1***
		(4)